

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$$7x + 6y = 5$$

$$28x + 24y = 20$$

For each real number  $r$ , which of the following points lies on the graph of each equation in the  $xy$ -plane for the given system?

Answer

A.  $\left(r, -\frac{6r}{7} + \frac{5}{7}\right)$

B.  $\left(r, \frac{7r}{6} + \frac{5}{6}\right)$

C.  $\left(\frac{r}{4} + 5, -\frac{r}{4} + 20\right)$

D.  $\left(-\frac{6r}{7} + \frac{5}{7}, r\right)$

Correct Answer: D

Rationale

Choice D is correct. Dividing each side of the second equation in the given system by 4 yields  $7x + 6y = 5$ . It follows that the two equations in the given system are equivalent and any point that lies on the graph of one equation will also lie on the graph of the other equation. Substituting  $r$  for  $y$  in the equation  $7x + 6y = 5$  yields  $7x + 6r = 5$ . Subtracting  $6r$  from each side of this equation yields  $7x = -6r + 5$ . Dividing each side of this equation by 7 yields  $x = -\frac{6r}{7} + \frac{5}{7}$ . Therefore, the point  $\left(-\frac{6r}{7} + \frac{5}{7}, r\right)$  lies on the graph of each equation in the  $xy$ -plane for each real number  $r$ .

Choice A is incorrect. Substituting  $r$  for  $x$  in the equation  $7x + 6y = 5$  yields  $7r + 6y = 5$ . Subtracting  $7r$  from each side of this equation yields  $6y = -7r + 5$ . Dividing each side of this equation by 6 yields  $y = -\frac{7r}{6} + \frac{5}{6}$ . Therefore, the point  $\left(r, -\frac{7r}{6} + \frac{5}{6}\right)$ , not the point  $\left(r, -\frac{6r}{7} + \frac{5}{7}\right)$ , lies on the graph of each equation.

Choice B is incorrect. Substituting  $r$  for  $x$  in the equation  $7x + 6y = 5$  yields  $7r + 6y = 5$ . Subtracting  $7r$  from each side of this equation yields  $6y = -7r + 5$ . Dividing each side of this equation by 6 yields  $y = -\frac{7r}{6} + \frac{5}{6}$ . Therefore, the point  $\left(r, -\frac{7r}{6} + \frac{5}{6}\right)$ , not the point  $\left(r, \frac{7r}{6} + \frac{5}{6}\right)$ , lies on the graph of each equation.

Choice C is incorrect. Substituting  $\frac{r}{4} + 5$  for  $x$  in the equation  $7x + 6y = 5$  yields  $7\left(\frac{r}{4} + 5\right) + 6y = 5$ , or  $\left(\frac{7r}{4} + 35\right) + 6y = 5$ . Subtracting  $\left(\frac{7r}{4} + 35\right)$  from each side of this equation yields  $6y = -\frac{7r}{4} - 35 + 5$ , or  $6y = -\frac{7r}{4} - 30$ . Dividing each side of this equation by 6 yields  $y = -\frac{7r}{24} - 5$ . Therefore, the point  $\left(\frac{r}{4} + 5, -\frac{7r}{24} - 5\right)$ , not the point  $\left(\frac{r}{4} + 5, -\frac{r}{4} + 20\right)$ , lies on the graph of each equation.